

5(a)(i)	<i>Mycobacterium, tuberculosis / bovis ;</i>	1
5(a)(ii)	<i>any three from</i> 1 phagocytes / macrophages / neutrophils / elastase, breaks down / weakens, alveolar walls ; 2 (reduced elastin so) alveoli cannot stretch and recoil / alveoli overstretch ; 3 alveoli burst / one large air sac formed ; 4 surface area (for gas exchange) is reduced ; 5 less oxygen absorbed by blood / reduced oxygen supply (to tissues) / not enough oxygen for (aerobic) respiration / AW ;	3
5(b)	R cost <i>any five from</i> 1 <i>Transmission ;</i> e.g. aerosol / droplet, infection I airborne A lack of pasteurised milk overcrowded, housing / living conditions / AW migration / travel, from areas with high rates of TB reservoir of infection in people not, diagnosed / not treated 2 <i>HIV / immunodeficiency ;</i> e.g. infection with HIV may activate (dormant) TB (pathogen) immunodeficiency / AW, makes people more susceptible to TB intravenous drug taking increases risk of, immunodeficiency / HIV 3 <i>Diagnosis is difficult ;</i> e.g. TB (pathogen) remains dormant / symptomless carriers limited access to health care for diagnosis / AW <i>idea that</i> contact tracing is difficult	5

Question	Answer	Marks
5(b)	4 <i>Treatment</i> ; e.g. treatment is long term / treatment may not start early enough people may not finish the course of treatment limited access to health care for treatment 5 <i>Acquired resistance</i> ; e.g. bacterium / TB (pathogen), develops resistance to, antibiotic(s) / drug(s) multi-drug resistance / MDR-TB / XDR-TB 6 <i>Vaccination</i> ; e.g. vaccine may not always be effective in providing protection difficult to achieve herd immunity concerns / misconceptions, about vaccination limited access to vaccination (programmes) poor (immune) response in people who are malnourished poor thermostability of vaccine 7 <i>Education</i> ; e.g. limited education / low awareness, about prevention for general population limited education / low awareness, for health care professionals	5